

Module 1: Transcendental Constant Definition

Certifies: $\alpha_0 = 299 + \pi/10$ to 5000 decimal digits

Machine Certificate v1.6 David Fox May 21, 2026

STATUS: CERTIFIED ✓

All SHA-256 digests verified. Source, stdout, and `invariants.json` bound.

1. Claim

The transcendental α_0 used in S_{14} is defined as $299 + \pi/10$ computed to 5000 decimal digits via `mpmath` arbitrary-precision floating-point arithmetic (`mp.dps = 5000`).

2. Source Code

File: `certificates/alpha0.py`

```
from mpmath import mp
mp.dps = 5000
alpha0 = 299 + mp.pi/10
print(alpha0)
```

3. Build Environment

```
$ python3 --version
Python 3.11.14

$ pip show mpmath | grep Version
Version: 1.3.0

$ sha256sum alpha0.py
8175b0e904084156c249ccc185420aa98db982976320247a54f082442b6d1d49
alpha0.py
```

4. Raw Execution Log

First 120 digits of stdout (full output: 5000 digits):

```
299.314159265358979323846264338327950288419716939937510582097494459230
781640628620899862803482534211706798214808651328230664709384460955058
2231725359408128481117450284102701938521105559644622948954930381964428
8109756659334461284756482337867831652712019091456485669234603486104543
2664821339360726024914127372458700660631558817488152092096282925409171
5364367892590360011330530548820466521384146951941511609433057270365759
591953092186117381932611793105118548074462379962749567351885752724...
```

5. Cryptographic Binding

All three SHA-256 digests are computed with `sha256sum` (SHA-2, 256-bit) over the exact bytes of each artifact.

Item	SHA-256 Digest
Source <code>alpha0.py</code>	8175b0e904084156c249ccc185420aa9 8db982976320247a54f082442b6d1d49
Stdout (alpha0 value)	63ef870a78766619327e99b68683bcef f8c8ef9a525298756c77c8378fd2c291
<code>invariants.json</code>	abe8cf160531964ff90371794c25fa8e cbc3330ceb434ca3d69cc6597f1cc235

6. Verification Status

Check	Result
$\alpha_0 = 299 + \pi/10$ computed to 5000 dps	PASS ✓
Source SHA-256 bound	PASS ✓
Stdout SHA-256 bound	PASS ✓
<code>invariants.json</code> SHA-256 bound	PASS ✓
Overall certificate status	VERIFIED ✓

Reproduce:

```
python3 certificates/alpha0.py | sha256sum
# should match: 63
ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291
```

Repository: <https://github.com/DavidFox998/alpha0-ponti> · Certificate generated by Machine Certificate v1.6